Evaluation of AI Techniques using Affective Computing Solutions within Video Games

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Abstract

Context: With video games becoming increasingly more realistic and powerful, with resources primarily focused on graphics and narrative. While AI systems and interactions lack behind in realism and can ruin immersion. Affective gaming could potentially improve this immersion and satisfaction of AI by incorporating the user's psychology and physiological data.

Aim: To develop a prototype system to analyse the effectiveness and accuracy of AI techniques with the use of affective sensors within video games.

Method: A prototype application was developed in the form of a horror game using the Unity game engine, making use of the Unity Asset Store for the audio and art assets. Once background research was complete into suitable AI techniques, fuzzy logic and dynamic scripting were chosen to be implemented. Further research went into the construct of the questionnaires to be used for both qualitative and quantitative data for the application.

Results: The results from participant testing have indicated that the fuzzy logic system performed overall better than the dynamic scripting. With all users selecting the fuzzy logic as their preferred system when testing. This is also confirmed from SAM questionnaire data, as well as promising signs from the emotional output that were calculated from the fuzzy logic.

Conclusion: Out of the 2 AI techniques tested, fuzzy logic seems to be the more effective system within the context of affective gaming. However, further testing with a greater range of participants is required.

Abbreviations, Symbols and Notation

NPC – Non-Playable Character

RBS – Rule Based System

AI – Artificial Intelligence

DOM – Degree of Membership

FIS – Fuzzy Inference System

SAM – Self-Assessment Manikin

1. Introduction

1.1 Overview

Over the past decades video games have become increasingly more powerful both in player experience and technology, with the development of graphics, AI, story narrative and gameplay to seem more realistic and engaging. With competitiveness within video games industry ever increasing, developers traditionally focus on graphics and narrative to attract more players (Lara-Cabrera et al. 2019). This focus can lead to AI/NPC interactions being less realistic or immersive. Affective games are an upcoming area within video games interpreting and assessing player's emotion to adapt gameplay, to improve their overall satisfaction and immersion.

This area of research comes under affective computing - also referred to as emotion AI or artificial emotional intelligence – which attempts to recognize, interpret and process human affect. For an affective computing system to detect emotional information it usually requires sensors to record data that will then be analysed by the AI system. Previous attempts have been made to use the players "emotional intensity" to adapt certain gameplay elements, in games such as Left 4 Dead 2 (Valve, 2009) and Alien: Isolation (Creative Assembly, 2014). While these games did not use affective sensors, the AI techniques – referred to as 'AI directors' (Thompson, 2017) – have the potential for further experimentation with affective games. More recent games such as Nevermind (Flying Mollusk, 2015) and Bring to Light (Red Meat Games, 2018), use affective sensors to track player's emotional state during run time to adapt gameplay elements of the game.

Affective sensors are a more elaborate approach to input devices for players to interact with the game (Kotsia et al. 2013). An AI system can then analyse data from these sensors and determine an appropriate strategy using affective models to adapt gameplay, this can be done via elements such as environment, Non-playable characters (NPC) interactions, game difficulty etc (Vachiratamporn et al. 2014).

Sensors directly measure physiological signals produced by the body. Types of measurements include skin temperature, heart rate, galvanic resistance, respiration, etc. Other sensors that identify emotional cues consist of, cameras that are capable of recording facial expressions, body posture, and gestures, whereas microphones could capture speech and its characteristics such as pitch, rate, rhythm, etc. The following project will aim to implement and use at least 3 different sensors to capture user data, this is due to the fact not all sensors are 100% accurate, so having multiple readings will increase the chance of a more definite outcome. Recognizing emotional information requires the data gathered to be analysed for sequences that can indicate human emotions, this is achieved with machine learning techniques, such as neural networks, fuzzy logic, fractal dimension or hidden Markov modelling (Guthier et al. 2016). In psychology one method of understanding and mapping human emotions are via models of affect (or affective models). The clearest method of representing emotions computationally is to label emotions into discrete categories. However, this can over-generalised, with discrete labels not taking into account the possible causes of emotion, mixed emotion or even a variety of emotions within the same category. A dimensional model is a more flexible approach for affective modelling, allowing each emotion to be represented as a point in a multi-dimensional model.

An affective model can be designed and implemented into a fuzzy inference system (FIS), which takes sensor input data (such as heart rate or blood pressure) and processes them based off the pre-determined rules of the affective model to map against emotional outputs, where both input and output are numerical values.

The emotional output from an FIS can be fed into a rulebased system. These systems are designed manually using domain-specific knowledge. An interface engine takes information from actions or events to a match-resolve-act cycle, which processes the input information to trigger a rule. Dynamic scripting (Spronck et al. 2004) is an unsupervised learning algorithm that makes use of rule bases. The probability of a rule being

selected and activated is proportional to a weight value associated with that rule. The rule base is updated via a reinforcement-learning technique, the weight values are changed depending on the success/failure rate with the corresponding rule. The output action from dynamic scripting will then adapt and alter gameplay elements as mentioned above, in hopes of varying user emotion.
The purpose of this project is to establish how effective AI decision-making techniques can improve the user's experience and immersion within a video game context. A basic horror game prototype will be tested by participants while wearing affective sensors, to determine if they noticed any change of events or gameplay elements based on their emotions/mood. This will be done via a questionnaire asking the user about their experience of the game, as well as their emotions/mood before and after testing.

1.2 Aim

To develop a prototype system to analyse the effectiveness and accuracy of AI techniques with the use of affective sensors within video games.

1.3 Objectives

Main objectives of the project are:

- Research current AI decision making algorithms and affective models.
- Determine which AI techniques could be implemented using affective sensors.
- Develop a first-person prototype horror game for the AI system to adapt and adjust.
- Produce an end survey for users, to attain qualitative feedback to assess the multiple affective sensors.
- Analyse and evaluate results for future improvements and recommendations.

1.4 Research Question

How effective are AI techniques with the use of affective sensors at improving player's experience and immersion?

1.5 Summary of Dissertation

Section 2 will consist of a literature review that will cover a comprehensive analysis into the current state of affective computing, AI techniques and past examples of affective computing within video games. Section 3 leads onto the methodology, which will detail the approach taken to construct the project. Section 4 goes into the results to cover the data recorded from the participant testing. Then finally section 5 will bring about the discussion and recommendations of future work.

2. Literature Review

This section will aim to provide an overview of the current state of affective computing, decision making AI techniques, and affective sensors. Covering the key elements of what affective computing is, how sensors play their part within this area of research, what AI techniques are currently used within affective systems and decision making in general, and how circumplex model/affective grids can be used to identify primary emotions through arousal and valence.

2.1 Affective Computing

As discussed within the introduction affective computing is an area of research which attempts to analysis human emotions through sensors, this field covers computer science, psychology, and cognitive science. One of the earliest mentions of affective computing comes from Picard, who proposed that computers could have the ability to recognize and express emotions (Picard, 2000). Due to this project being based within the context of video games, the methods researched and discussed here will be related to this – sometimes referred to as affective gaming. According to Hudlicka there are 3 core areas of affective computing which can provide methods and techniques directly relevant to game development, consisting of: 1) Emotion sensing and recognition by machines; 2) Computational models of emotion; and 3) Expressive manifestations of emotions, and effects on behaviour, problem-solving, and decision-making (Hudlicka et al. 2009). This project will primary focus on the first area of interest. Affective computing systems require sensors to read data from, which can then be processed to indicate what emotion the user is currently feeling. There are a wide range of sensors from heart rate, galvanic skin response, webcams, microphones, and electromyography. It is important to note however, that in the context of games there is no optimal sensing technology, as it differs for each individual game (Guthier et al. 2016). For the system to identify emotions there will need to be measurements of both arousal and valence, meaning a minimum of 2 sensors will be required, this will also allow the

observation of the recorded data to ensure that both have similar trends when analysing emotions. The project will test the game with participants to evaluate the effectiveness of the AI system. Due to their being multiple participants, data from input sensors will be required to be normalized as there are large differences in individuals associated with physiological data (Mandryk et al. 2007).

2.2 Affect Grids

This section examines the psychology of emotions and the existing methods to measure and record emotions. Emotion has been well researched field of psychology for decades being thoroughly understood subject. It is important to understand the current theories related to the modelling of emotions and how developers can then construct AI systems which will analyse emotions. From early studies within the field of discrete emotion theory, it was determined that there are 6 basic emotions consisting of anger, disgust, fear, happiness, sadness and surprise no matter the cultural background of an individual (Ekman 1992). This is meaningful information as not only are these emotions cross-culturally recognisable, but they can be characterised as ‘discrete’ meaning they are believed to be detectable by an individual facial expressions and biological processes. When it comes to modelling emotions there has been many methods of computationally representing emotions in AI systems however, opinions differ on whether discrete labelling of emotions vs a multidimension model of emotions is a better representation of emotions, both with limited success of identifying emotion states (Mandryk et al. 2007). The easiest way of representing emotions within an affective model would be to label them into discrete categories. However, this method can over-generalize this down to a small number of categories, which also does not take into account the possible cause of emotions, mixed emotions or even a variety of emotions within the same category (Guthier et al. 2016). An alternative affective model to consider would be a circumplex model of emotion by Russell et al. (1989), in which emotions are distributed in a 2D-circular

space, based off arousal and valence dimensions, arousal being the amount of physiological change that the emotion causes a person, and valence identifies between pleasant and unpleasant emotions, this alternative model being called an affect grid (Figure 1).

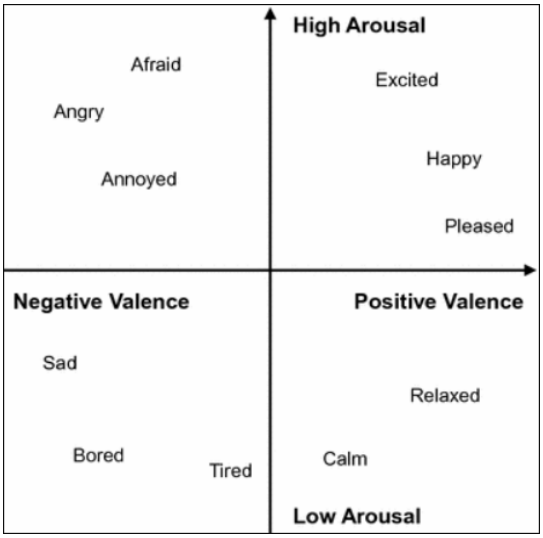

Figure 1 – Two-dimensional affective model (Russel et al. 1989)

A dimensional model is a more flexible approach for affective modelling, allowing each emotion to be represented as point in a multidimensional model (Guthier et al. 2016). This two-dimensional model can be further expanded to three dimensions with the third dimension, dominance, also referred to as a model of pleasure, arousal and dominance (PAD). This third dimension can be described as feeling of control and exerting influence over others vs feeling of weakness and submission to external influences (Guthier et al. 2016). For the size and scope of this project, a two-dimensional affect grid was chosen as the appropriate method, this is due to the labelling method being potential too simple, while not as complex as a three-dimensional affect grid either, thus being the selected affect grid method.

2.3 Fuzzy Logic

To process the sensor data AI techniques can be introduced to analyse and interpret into categorized human emotions. Fuzzy logic is an AI technique which offers ‘degrees of truth’ rather than the conventional 1s and 0s (Boolean logic) represented into computer logic. This approach

first came about by Zadeh (1965), which attempts to mimic human logic through natural language containing uncertainties or vagueness, with words such as quite, very, relatively, etc. This is done by appointing a value between 0 and 1, this process is called fuzzification, where computers take a crisp value and interpret linguistic rules through set operations, before they can then be defuzzified back into a crisp value to be outputted, this is also known as fuzzy rule-based inference (Buckland, 2005).

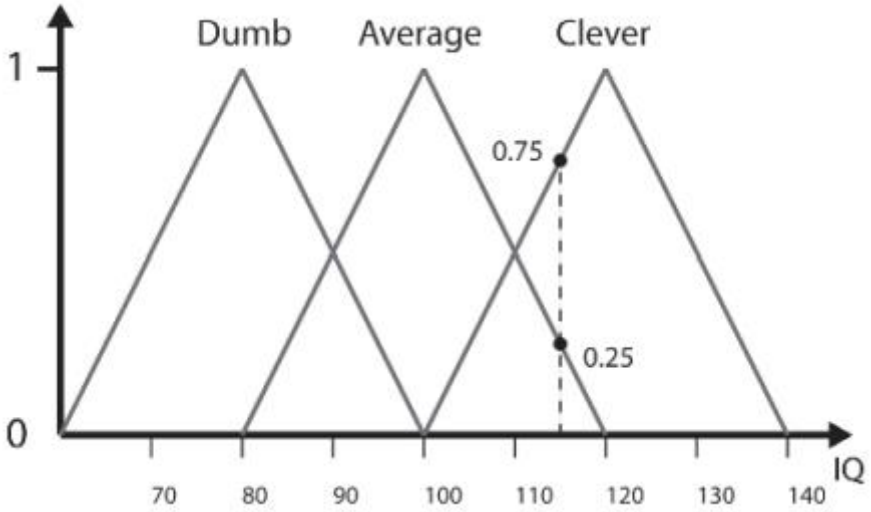

Figure 2 – Fuzzy sets representing intelligence through fuzzy logic (Buckland, 2005)

Figure 2 gives an example how of intelligence can be represented through fuzzy sets each containing a membership function, these functions make up the universe of values within a set, while the membership function can be any arbitrary shape, common shapes include: triangular, trapezoidal, or gaussian. The construction the rule base required for an FIS is usually done by an expert, who would create rules commonly through IF/THEN statements with set operations such as AND, OR, NOT. Defuzzification is the reverse process of fuzzification, which takes fuzzy outputs from the fuzzy set's membership functions and converts then into crisp output of the system (Shukla et al. 2010). There are many methods for defuzzifying process, some of the most common include mean of maxia, average of maxima, and centroid. For this project average of maxima approach was chosen for defuzzification stage, this is

part due to the ease of implementation of code while still being quite accurate in output. One of the main issues with FIS is that once the system starts to increase in complexity, the number of rules grows rapidly, with the equation for the number of rules needed to be constructed seen in Figure 3.

$$\text{Num. of rules} = \text{Num. of MFs per variable}^{\text{Num. of variables}}$$

Figure 3 – Equation for the number of rules needed for a FIS

So, a FIS which has 3 variables containing 5 membership functions would account to 125 rules. To reduce this rapid increase of rules, an alternatively method called Combs can be introduced to decrease the number of rules from exponential to a linear expansion (Combs, 1997). Following the example earlier of 3 variables and 5 membership functions, Combs method would reduce the rulebase from 125 to 15, this would dramatically increase computation time, especially as the originally rulebase grows exponentially. Depending on the design and size of the FIS this method may be used as an alternative to the traditional rulebase method, to act as a comparison.

2.4 Dynamic Scripting

Game difficulty adjustment more commonly known as dynamic difficulty adjustment (DDA) is a method of modifying gameplay features and behaviours based off the player's current skill. The purposed of DDA within video games is to adapt the present challenge of the game, so that it similar to that of the player's skill, creating a more enjoyable experience for the player. Dynamic scripting (Spronck et al. 2004) is a novel decision-making algorithm which automatically optimises game AI during runtime. This technique is an unsupervised decision-making algorithm primary focused on game AI, the 2 fundamental components that dynamic scripting requires to function are a rulebase and weight system. First a rulebase needs to be manually designed using domain-specific knowledge, in the context of this project these rules will act as the conditions required to trigger actions that the AI should take to scare or calm a user depending on the parameters passed into the algorithm. The

second crucial element needed is a weight system for the rulebase, this involves each rule having a correlated weight value. These weight values are initially set to zero, after the AI processes the rulebase the weight values are changed depending on the success or failure rate linked to the rule triggered. The probability of a certain rule from the rulebase being triggered is determined by the weight value linked to the rule, the higher the weight value the more likely it will be triggered, figure 4 shows an overview of the dynamic scripting process through the context of a fighting game.

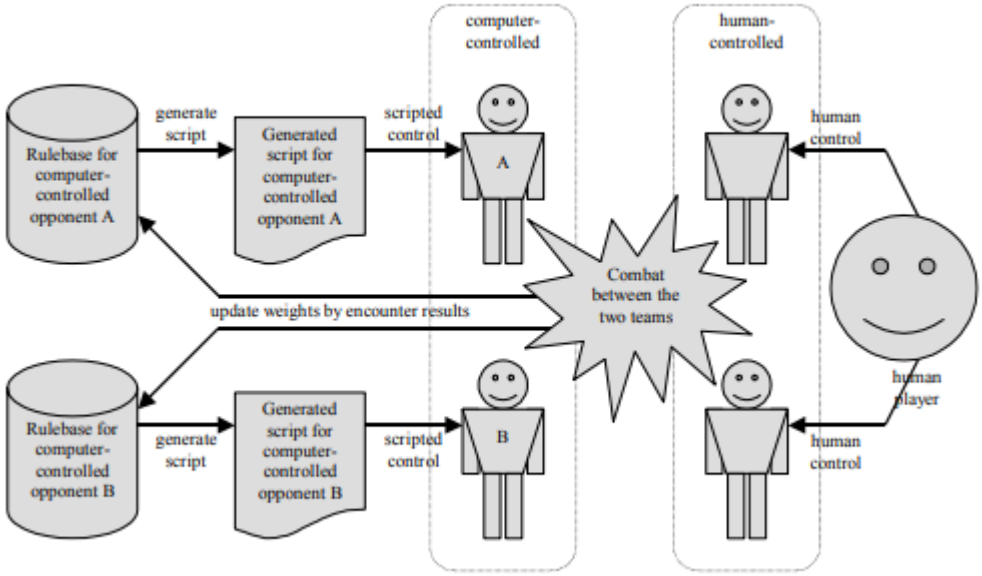

Figure 4 – Dynamic scripting process

(Spronck et al. 2004)

The weight adaption process was inspired by reinforcement-learning techniques (Russell and Norvig 2002), however was altered due to traditional reinforcement-learning techniques not being suitable for games. Once the process of the AI is complete the weight value of each rules is adjusted depending upon their success or failure, with the remaining rules being adjusted so that the total value of the weights remains the same through the rulebase.

2.5 Summary

In summary, there are various ways of implementing an affective AI system within the context of video games – from different AI techniques, sensors for data input and psychological methods. In addition to the 3 core areas in which affective computing can relate to video games. This project will implement 2 different AI techniques – Fuzzy logic and dynamic scripting – whilst focusing one the first area of the 3 core areas of affective gaming, emotion sensing and recognition by machines. These AI techniques were chosen due to their effectiveness for decision making algorithms, while allowing the creation of rules via a rulebase simplistic and easy to implement via domain-specific knowledge. Affect grids allow for a clear visualisation of how emotions can be mapped and then translated into an FIS to then be processed to crisp integer values. The methodology covered in the next section will detail how these techniques were implemented.

3. Methodology

This section will detail the process taken to create and design a basic prototype game for testing the effectiveness of the AI algorithms. How the hardware and it's required software was chosen and integrated into the application. The construction and use of affect grids to help analyse emotions. Finally, the implementation of the required AI algorithms and the software design behind each one.

3.1 Game Development

For the overall application to work, a basic game is required to test the chosen algorithms. The genre of the game is an important aspect of how the designing phase of the AI is approached, while certain genres of games are great at getting the user to express emotions through AI interactions, others are limited in this aspect. For this application, a horror game was chosen as a suitable option to test the AI systems. This is due to elements of horror games being simple to implement through visual and sound aspects, with emotions associated with the horror genre being clear to measure. Unity (Unity, 2019) game engine was chosen as suitable software for this application due to prior experience and its wide range of functionality. The game consists of the user exploring a haunted house to find 5 different keys spawned randomly around the level to escape – figure 5 displaying the final level design of the game.

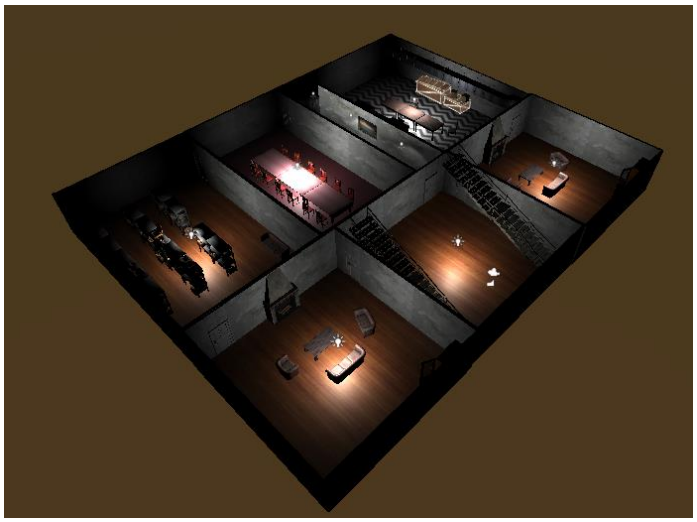

Figure 5 – Layout of game level

To keep the primary focus of the application on the 2 AI systems, the game was kept simple for more time to be allocated to the implementation of AI systems and making it easy for the user to play the game. Art and audio assets from the Unity asset store were utilised to create a more convincing haunted house environment for the user while saving development time instead of creating assets from nothing.

3.2 Hardware and sensors

The process of finding potential sensors for this project took a significant amount of time, this was due to the criteria of appropriate sensors being quite specific. First, a minimum of 2 sensors would be required overall for the application to function, these sensors need to measure the valence and arousal of the user. The second key element when selecting the correct sensors is the compatibility with Unity and being able to send the user's physiological data, another aspect to consider is the latency between the device and game. Finally, the affordability to the user as the average user cannot afford medical grade equipment worth potentially thousands of pounds. After exploring multiple options, including hardware offered by the university, the Seeed Grove (Seeed, no date) ear-clip heart rate sensor, and any standard webcam with the use of the Affectiva (Affectiva, no date) plugin were chosen. These sensors met the criteria needed for the application and are relatively straight forward to implement. Arousal being measured by the user's heart rate and valence being measured via the user's facial expressions.

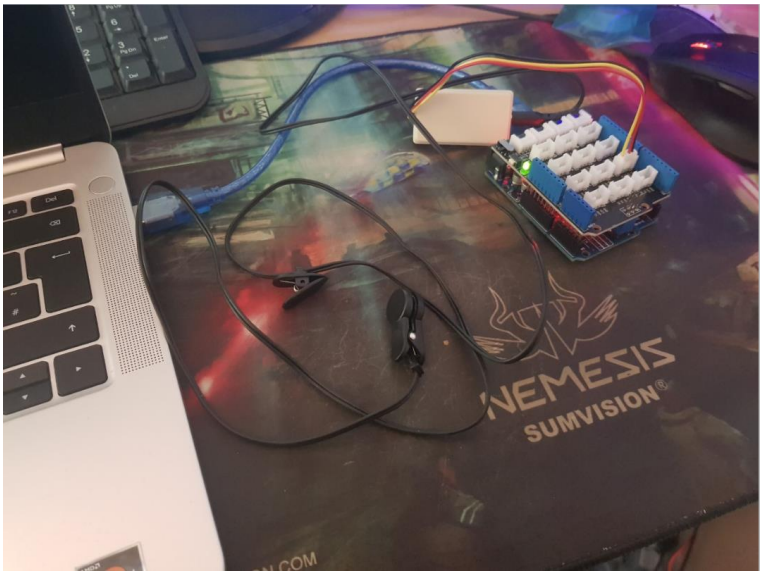

Figure 6 – Seeed Grove heart rate sensor with Arduino

Figure 6 displays the layout of the ear-clip sensor with the Arduino. The Seeed Grove ear-clip works from photoplethysmography, this is the method of detecting blood volume changes via infrared. The sensor can be used on the finger or via the earlobe, for this sensor it the most optical while placed on the earlobe according to the documentation. For the sensor to function a micro-controller is needed, for this application an Arduino board was chosen with the use of the Grove-base shield to allow input into the game. The script which calculates the heart rate on the Arduino updates every 5 seconds, this is the fastest it can update the heart rate input without losing significant accuracy. As mentioned before latency was one of the factors to be considered when choosing the desired sensor, which was a common issue among a lot of the researched sensors, some requiring the sensor data transferred via Bluetooth to a phone app to then be transferred again via WIFI. The result from these numerous stages is high latency of data from the user's live measurements to in-game. With the use of the micro-controller along with the sensor, the only transferred of data required is between the PC running the application and the micro-controller itself recoding the data. To ensure latency was as low as possible for runtime a custom Unity library called WRMHL (WRMHL, 2017) was used. This library allows for faster transfer of data over the conventional StreamReader class within

the .NET framework, with the application reading data from the WRMHL queue during runtime.
The second sensor chosen was a webcam with the use of the Affectiva Unity plugin for emotion measurements, the plugin recognizes human emotions based on facial cues or physiological responses. As Affectiva has its own Unity plugin implementation into the application was relatively straight forward, with some examples and documentation on how the plugin works. For the webcam to work and register human faces within Unity 2 Affectiva scripts are required, the Camera Input and Detector script. Camera Input script is primary for webcam input, the resolution of the image captured, and the sample rate of the webcam. The Detector scripted is used for selecting the appropriate emotions and expressions, for this project only the valence is required from the webcam input. Due to the valence output from the Affectiva script being on a scale from -100 to 100, this was altered to 0 to 100 for the fuzzy logic and dynamic scripting input which will be covered later within the methodology.

3.3 Fuzzy Logic

When originally designing the overall AI system for the application, the fuzzy interface system (FIS) was to manage the task of categorizing emotions based on arousal and valence inputs, with the use of affect grids. The dynamic scripting would then act as the rule base for triggering game-based rules from the input given from the FIS. However, after further research and discussions, this changed so that the FIS would become one sperate system to be compared against dynamic scripting. Now the fuzzy logic system would incorporate a basic rule base to execute game actions based on the relevant rules set in place.
The first major step to complete was the implementation of core fuzzy logic elements, this consists of Fuzzy rule, set, variable, and term. These scripts allow for the foundation of a FIS, allowing for the basic operations needed for fuzzy logic. Another critical requirement for fuzzy logic is membership functions, these functions can be any arbitrary shape however, some typical examples include triangular, trapezoidal, and

gaussian curve. Due to membership functions being represented by arbitrary shapes, Unity's AnimationCurve was the ideal system to use for this. The AnimationCurve allows for any shape to be constructed along a two-dimensional axis (time and value). Also, included is an evaluation function within the AnimationCurve class allowing for a value to be determined based on the time at a certain point, this value can be used as the degree of membership (DOM). The final aspect needed for fuzzy logic is a defuzzification method, which is the reverse process of fuzzification by turning a fuzzy set into a crisp value. There are many techniques for this process from centroid, mean of maximum, bisector of area, etc. For this application, the average of maxima was chosen, while this may not be the most accurate method – centroid with most the accurate output, yet most complex to calculate – this method is relatively simple to calculate with outputs close to that of the centroid method. Appendix A shows the finished overall structure diagram of how the fuzzy logic is constructed within the code.

Once these functions were achieved a simple fuzzy system was created to ensure that these base scripts functioned correctly. To ensure that the system worked MATLAB's Fuzzy Logic Toolbox (MATLAB, no date) was used to test the created system. From the testing, it was found that the output from the application's system compared to that of the MATLAB's was a percentage difference of 34.89%. It should be noted, however that during the testing MATLAB made use of the centroid defuzzification method, due to it not having the average of maxima method.

Now that the basic requirements of the FIS have implemented, the next stage was to choose and design the input and output membership functions. Due to having heart rate as an input into the system, 2 FIS were needed to translate this into a crisp emotional output value, as heart rate cannot be directly translated into a crisp emotional value by itself. The first FIS will convert heart rate into an arousal output, which can then be used as an input - alongside the valence reading from Affectiva - into the second FIS then modelling emotions from arousal-valence space. Before raw heart rate data can be inputted into the FIS it must first be

normalized, this is necessary when performing group testing, as individuals differ in physiological data. The following formula – as seen in figure 7 is to calculate the normalized heart rate data, which is outputted on a percentage level from 0 to 100.

$$Normalized\ HR(i) = \left(\frac{\mathrm{HR}(i) - HR_{min}}{\mathrm{HR}_{max} - \mathrm{HR}_{min}}\right) \times\ 100$$

Figure 7 – Formula for normalized heart rate

For the first FIS to calculate the arousal from the heart rate, 2 fuzzy variables are used to represent the input and output, each with 4 fuzzy sets that range from low reading to high reading. The rules used are relatively simple, consisting of 4 rules which can be found in Appendix B.1 with a linear projection of heart rate to arousal.

The second FIS was more complicated to construct, taking 2 inputs (arousal and valence) and 4 outputs (fear, calm, sad, excited). Originally a total of 6 emotions were going to be calculated, each taking a unique position within the affect grid – see figure 1. However, the decision was made to remove 2 emotions of alarmed and tiredness, as these both created overlap with existing emotions which were identified to be more important and would make the process cleaner to implement. The membership functions and rules for converting arousal and valence into emotions were created using affect grids. The arousal and valence were mapped into the 4 chosen emotions via a 5 by 5 divided grid, with each section representing the emotional levels (low, mid-low, medium, mid-high, high). Thus, 5 evenly distributed membership functions used for inputs of arousal and valence. From earlier research, it seems that there is no confirmed method of defining emotions from arousal-valence space, rather estimations and adjustment to fit the desired application. This application will make use of the ‘low’ level term as little to no emotion, scaling to high being extreme and obvious signs of emotion.

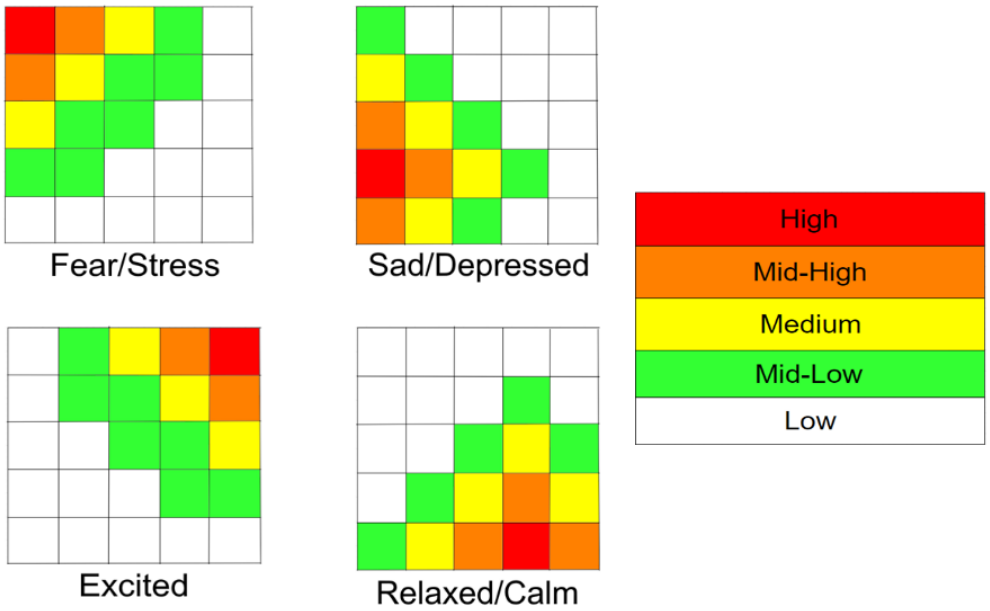

Figure 8 – Affect grids of selected emotions

Figure 8 shows the constructed affect grids used for this application, these grids allow for easy visualization of how arousal and valence can be mapped to emotions. These grids also allow for straightforward creation of the rules needed for each emotion, with 25 rules for each grid – See Appendix B.2 to view all rule tables.

For the designed rules (see appendix C) to be triggered within code, IF/THEN statements were used which resemble the same function of that with the rulebase. However, with multiple IF/THEN statements in code, it becomes sequential triggering the first condition which matches the parameters. To counter the potential issue of repeating rules over and over, each possible rule that can trigger is added to a list in which a random rule is chosen to be activated. This process starts after 20 seconds from when the game begins to give the user time to adjust to the game, with all subsequent rules are given a 20-second window after each triggered action so the user is not bombarded with actions.

3.4 Dynamic Scripting

The construction of dynamic scripting was relatively straightforward, due to the original authors of the algorithm publishing pseudocode for the technique alongside an explanation of the process. The algorithm

requires 2 essential components to function, firstly a rulebase to store the actions that the AI will trigger when certain conditions are met, and secondly a weight adjustment function to reorder the priority of rules based on their performance. The initial stage was the task of how to create a rulebase that is flexible in design. For every rule created a condition and action is required, this originally turned out to be an awkward process to implement at first, with traditional IF/THEN statements requiring that a new rule script be created for each rule so that the condition is hardcoded as well as the action. Eventually, a better solution came about, with delegate functions that carry lambda expressions. Appendix D shows the overall structure to dynamic scripting, with all rule conditions and action hardcoded via a central rulebase now, rather than having multiple rules scripts. This allows for a single list object of rules to be created to store all potential rules needed, with a designer of the rulebase simply creating with the conditions and actions of each rule via delegates.

The script generation and weight adjustment are called before and after each interaction the AI has with the user. This algorithm still makes use of the same 20 second grace period applied (as the same with the fuzzy logic system) at the beginning of the game, also with the same 20 second rule trigger time therefore after. However, the difference with this algorithm is that 10 seconds after a rule is triggered, a fitness value is calculated to determine if the rule was successful or not based on the user's physiological data, which is then fed back into the rulebase via the weight adjustment script. So, before an interaction begins, the script generator creates a new script to hold a certain number of rules to be tested for the upcoming interaction. The algorithm will aim to insert several rules to the newly created script but may end up with less so, a minimum number of rules is guaranteed to ensure the script has something to attempt to execute. The selection of rules is based on a SoftMax selection function, where the larger the rule weight will correspond to a larger probability of being added to the new script. To calculate the fitness a function is used after an interaction has occurred.

Which takes the heart rate and valence before and after the interaction to calculate the overall fitness, where the greater increase in heart rate and decrease in valence creates a positive fitness value and vice versa.

$$\Delta W = \begin{cases} -\left\lfloor P_{max}\frac{b-F}{1-b} \div 0.5\right\rfloor & \{F < b\} \\ \left\lfloor R_{max}\frac{F-b}{1-b}\right\rfloor & \{F \geq b\} \end{cases}$$

Figure 9 – Weight adjustment equation

Figure 9 displays the weight adjustment equation used for weight calculation after an interaction. The new weight for a rule is W + ΔW, where W is the original weight value. This equation is modified version from author's original due to the fitness value (represented by F within the equation) having a different range, with the author's fitness value based within the range of 0 to 1, while this application makes use of a range of -1 to 1. The break-even value b was set to 0.01 so that b was between the average penalised and rewarded values (which ranged from about -0.1 to 0.12). Finally, R_{max} and P_{max} serve as the maximum reward and maximum penalty values, based on the author's recommendations the R_{max} set to 100 for increased efficiency to allow large weight increases for the rules, while P_{max} was set to 70.

Once the fitness value and weight adjustment values are calculated, the rules from the used script are looped through, with adjustments made to any rule activated during the interaction with the appropriate updated weight value based on the weight adjustment equation.

3.5 Questionnaire Design & Testing Methods

For each participant testing the following data was recorded: sad, fear, excited, calm outputs along with heart rate and valence measurements. These quantitative measurements allowed for each AI technique to be assessed based on data provided via the user's physiological data. Such as how effective the overall structure of the rules was, depending on how much of an increase or decrease from the data. Also, to be counted throughout the testing was the number of rules triggered and the number of unique rules triggered for both techniques. This allowed to analysis

how consistent the techniques were across a wide range of different users.

Further quantitative data that was collected through questionnaires to estimate current user mood and emotions, these were filled out before and after testing each AI system with a 5-minute rest between testing each system to allow the body to rest. Originally the questionnaire was to consist of a profile of mood states (POMS) and a self-assessment manikin (SAM), both allowing to measure the user's emotions via images and word association. It was chosen that POMS may not be suitable for this type of testing. This was for many reasons starting with the type of emotions and moods that the user would be asked to measure, many not being relative or appropriate to the nature of the game/environment. Second, the sheer volume of questions required to complete each questionnaire, due to it needed to be filled out 4 times in total. The original POMS questionnaire totalled 65 items, while a shorter version of this questionnaire was cut to 35 items, this would still require the user to fill out a total of 140 items.

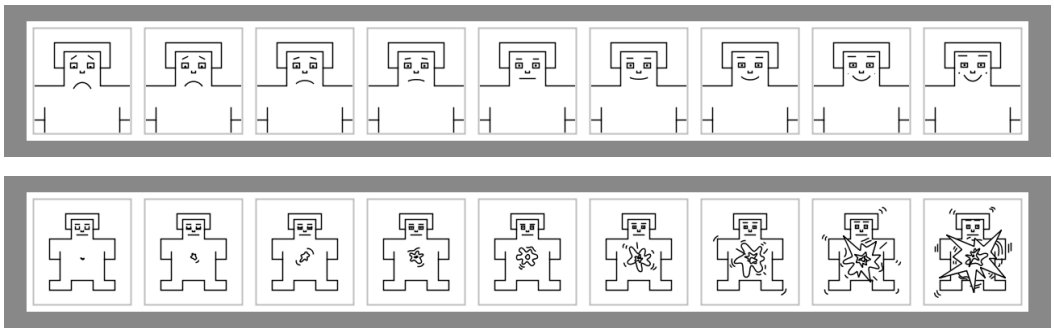

Figure 10 – Example of SAM questionnaire

While a SAM (figure 10) questionnaire is straightforward, with the user selecting which imagine represents their feelings. Each of the rows of images representing valence and arousal on a scale of 0 to 9.

Finally, a qualitative questionnaire was created to gain overall feedback about both AI techniques, and which one was preferred, if the user noticed any of the game events based off their emotions, if the equipment was intrusive at all, and if they could see affective gaming having a future within the video games industry.

4. Results

Due to the current circumstances which have restricted testing, the testing of the application only consisted of 3 participants. This will be explained further within the next section.

The participants were asked to fill out the SAM survey before and after testing each system, this would give further quantitative on top of the physiological data recorded. Before starting each test of the AI systems, the participant's heart rate was measured for 1 minute to calculate their normalized heart rate. After testing both of the AI systems the participants were asked to fill out the end of testing questionnaire – exact questions are displayed in figure 11. These questions were aimed to gather qualitative feedback about both AI systems as well as user's opinions on affective gaming, these were answered with closed-ended answers.

Q1)	Do you personally watch, read or ply games within the horror genre?
Q2)	Out of the 2 demos tested did you prefer the first or second?
Q3)	Did you notice any in-game events (such as audio playing, lights turning off and on, or jump scares) in correlation to your emotions?
Q4)	Did you find any of the equipment intrusive at all?
Q5)	Do you believe affective gaming has a future within the video games industry?

Figure 11 – End of testing questionnaire

The following will summarise the presented data that was taken during testing. The full responses to the end of testing questionnaire can be found in Appendix E. Question 1 was intended to find out if the user interesting in the horror genre which would affect their opinion on the game, the majority answering no. The rest of the question was aimed at gaining feedback on the game and systems, starting with preference in which AI technique with 100% response to the fuzzy logic system. This 100% response continued in the last 3 questions when asking feedback

about the in-game events, the intrusiveness of equipment, and the potential future of affective gaming, all being positive.

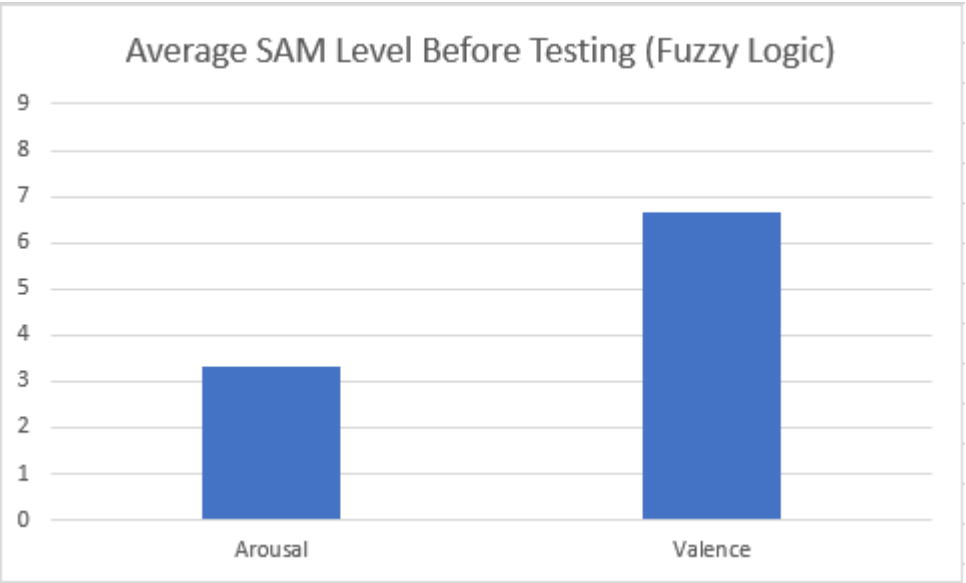

Figure 12 – Average Levels of arousal and valence before testing (Fuzzy Logic)

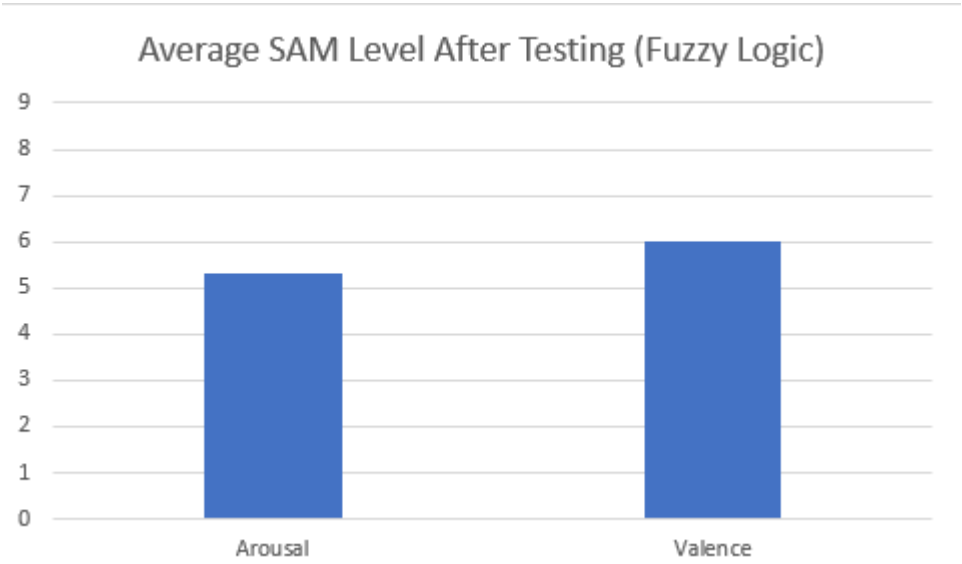

Figure 13 – Average Levels of arousal and valence after testing (Fuzzy Logic)

Figures 12 and 13 display the average outcome of the SAM survey given to the participants before and average testing the fuzzy logic AI. With an overall increase in arousal and a decrease in valence, which indicates a higher heart rate and an increase in unpleasant emotions.

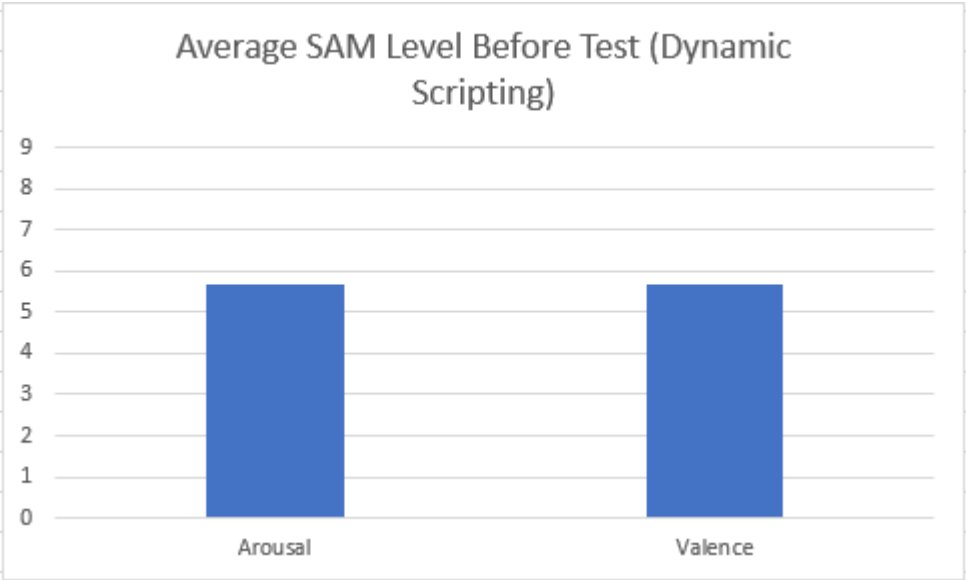

Figure 14 – Average Levels of arousal and valence before testing (Dynamic Scripting)

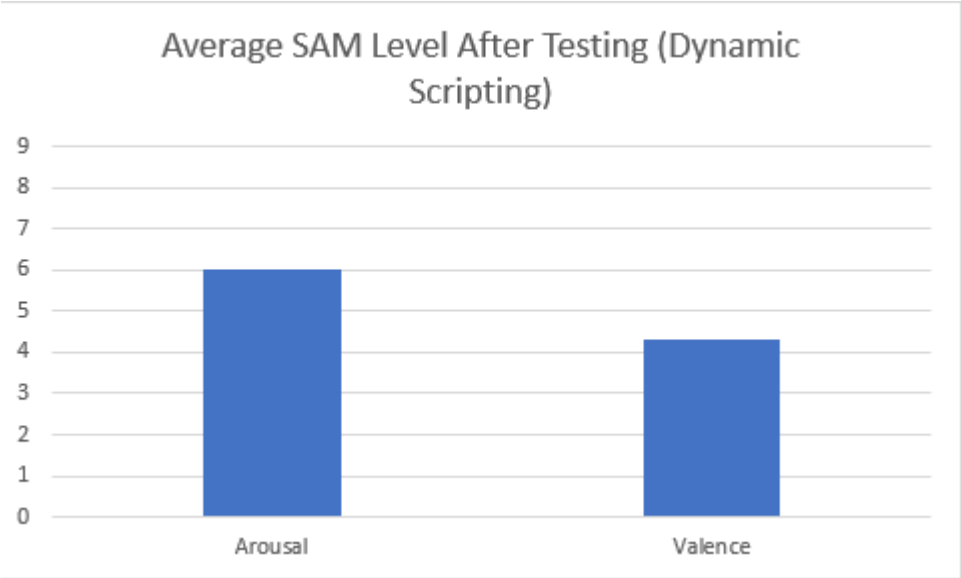

Figure 15 – Average Levels of arousal and valence after testing (Dynamic Scripting)

Figures 14 and 15 show the same SAM survey being answer by participants before and after testing the dynamic scripting AI. A similar trend can be seen to that of the fuzzy logic system, but a lesser extent.

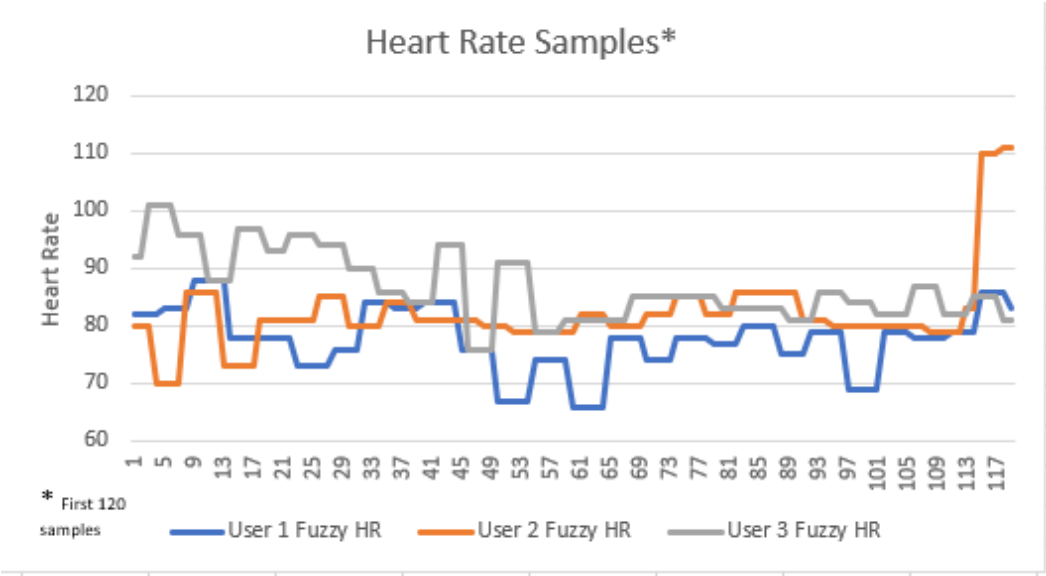

Figure 16 – Fuzzy logic heart rate samples

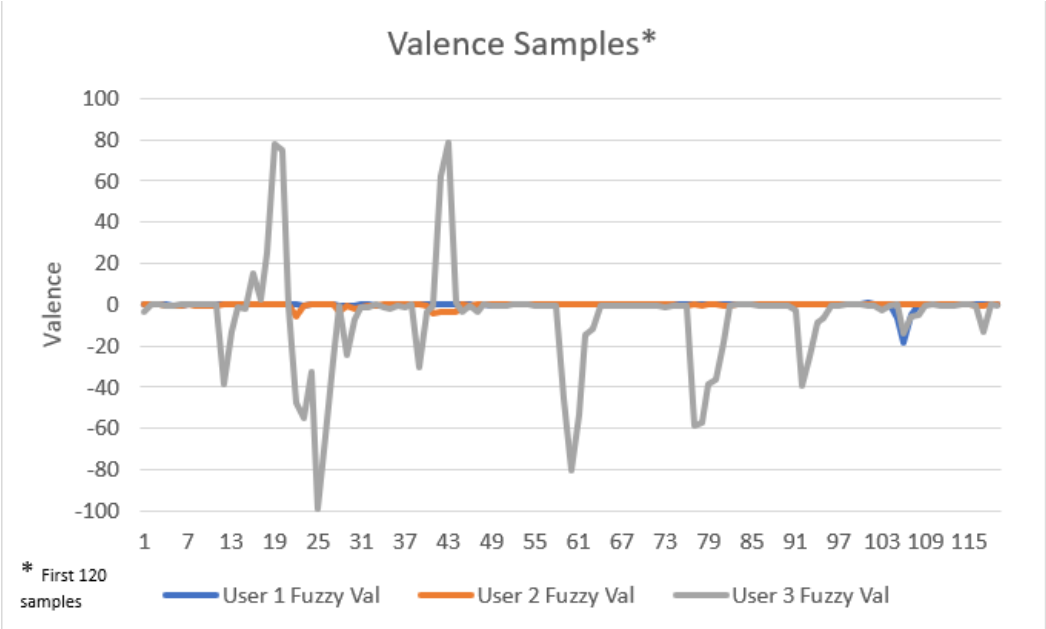

Figure 17 – Fuzzy logic valence samples

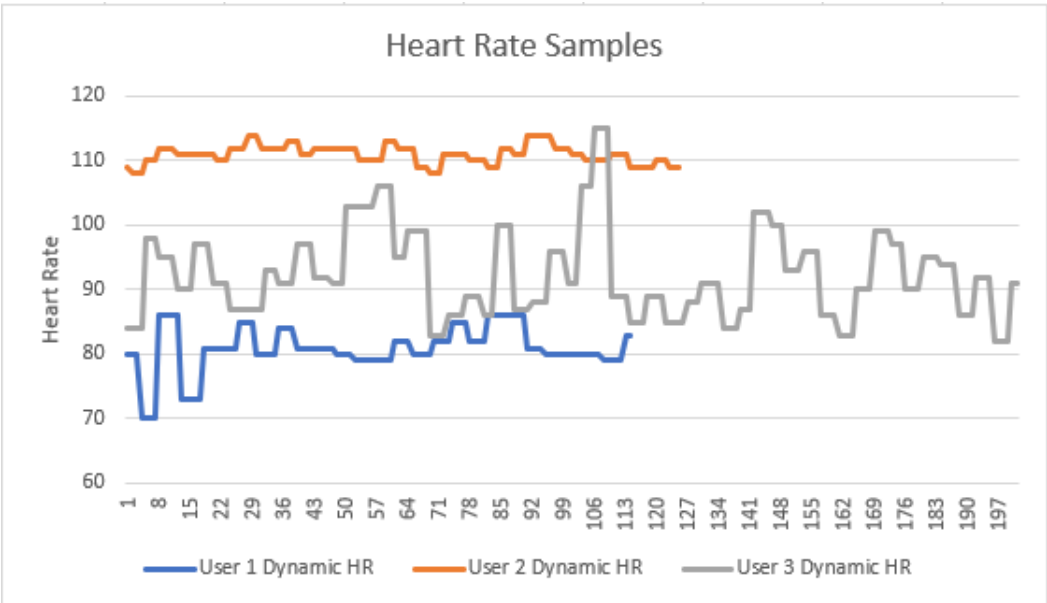

Figure 18 – Dynamic scripting heart rate samples

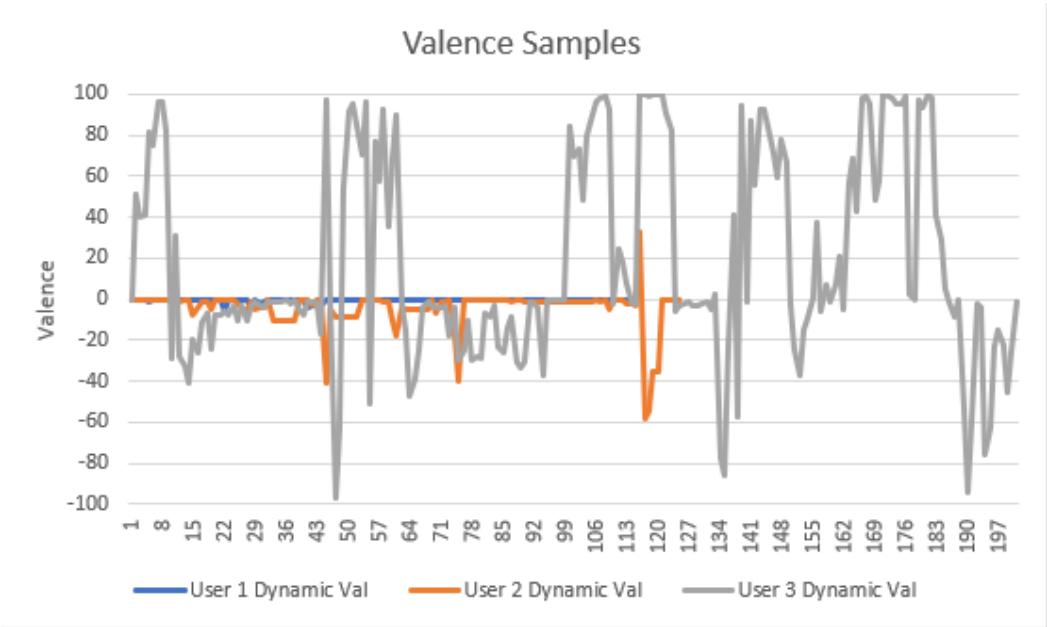

Figure 19 – Dynamic scripting valence samples

The graphs from figures 16 – 19 show the raw heart rate and valence data that was recorded during each testing of the AI systems. Some points to identify, the reason user 2's heart rate and valence were altered was due to the programming recording data sometime after the end of the game. Also, the length of each line within the graphs indicates the user's total playtime of the game with the specification AI system.

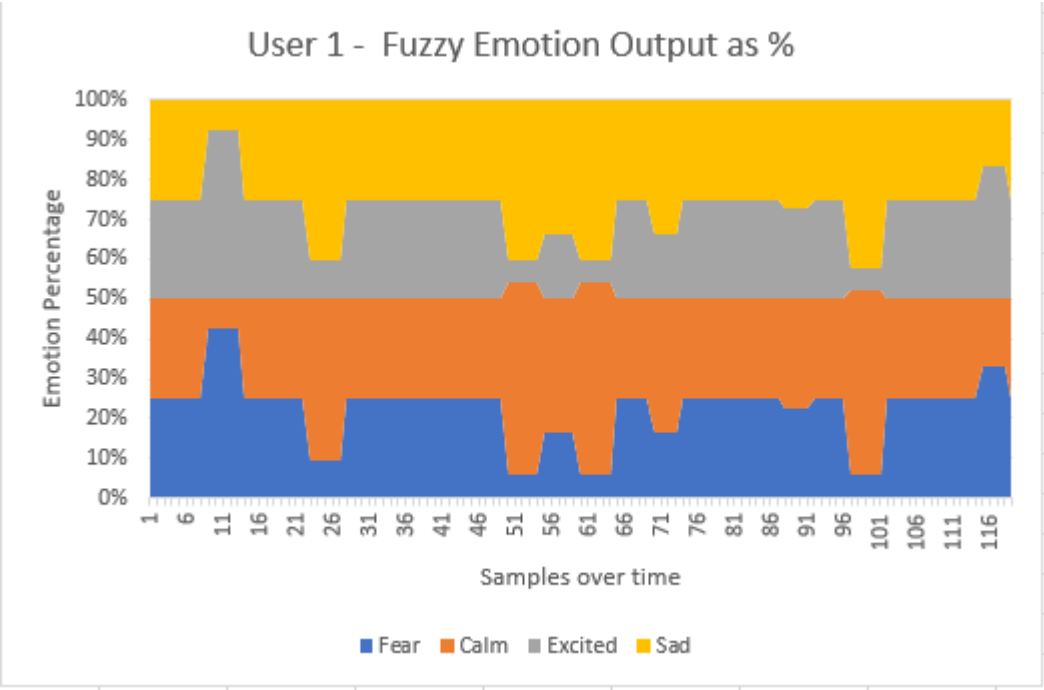

Figure 20 – User 1 fuzzy logic emotions as a percentage

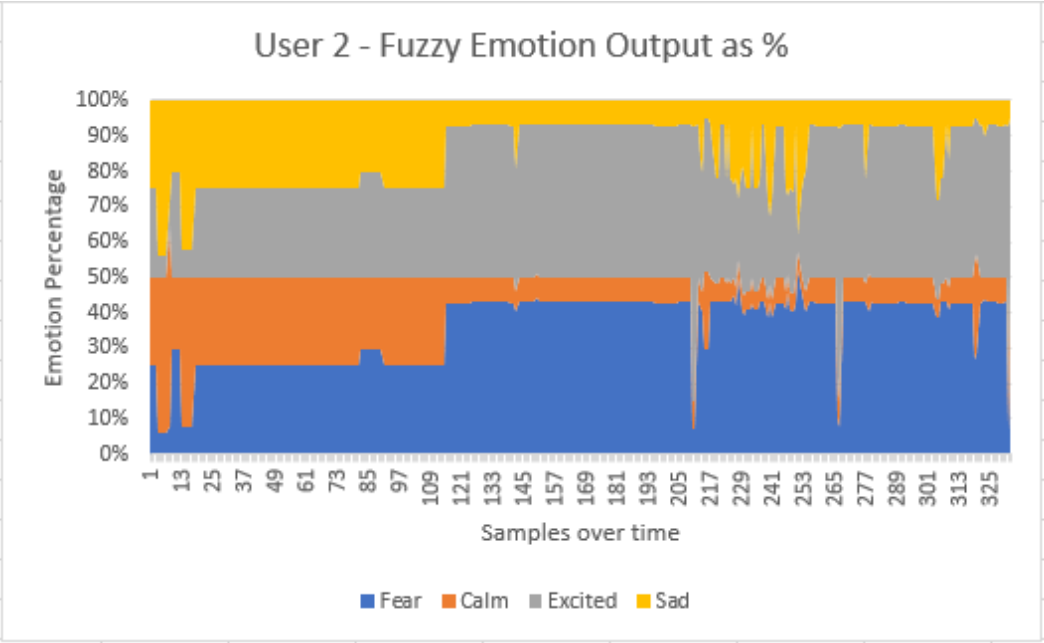

Figure 21 – User 2 fuzzy logic emotions as a percentage

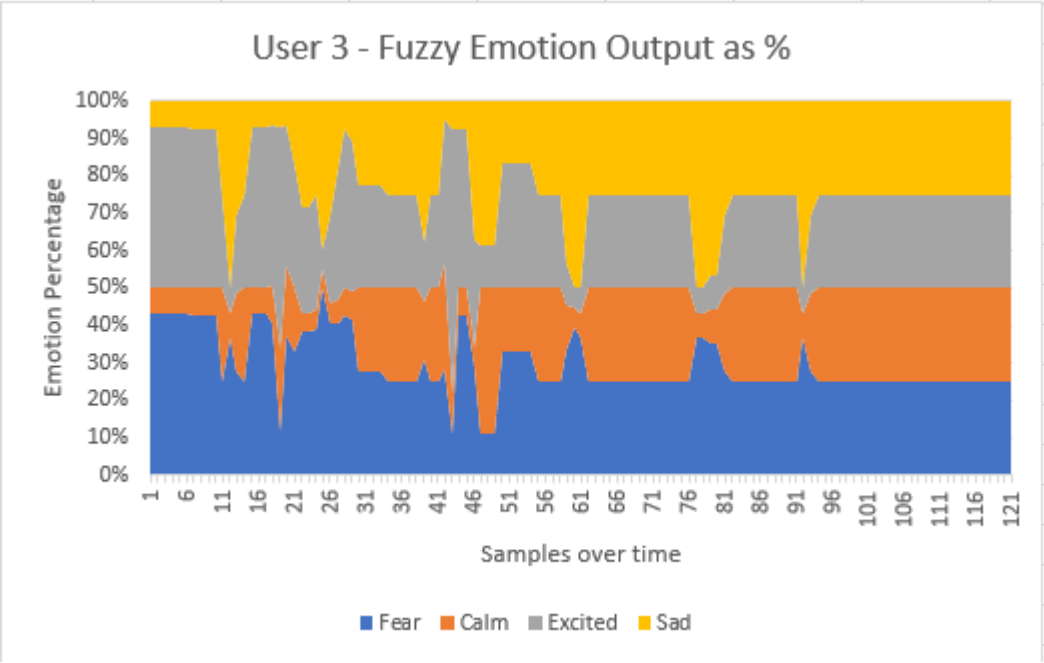

Figure 22 – User 3 fuzzy logic emotions as a percentage

The stacked area charts from figures 20 to 22, display each user's trend of fuzzy emotions outputted throughout the game. The greater vertical area of the emotion within the chart indicates a higher output of the emotion.

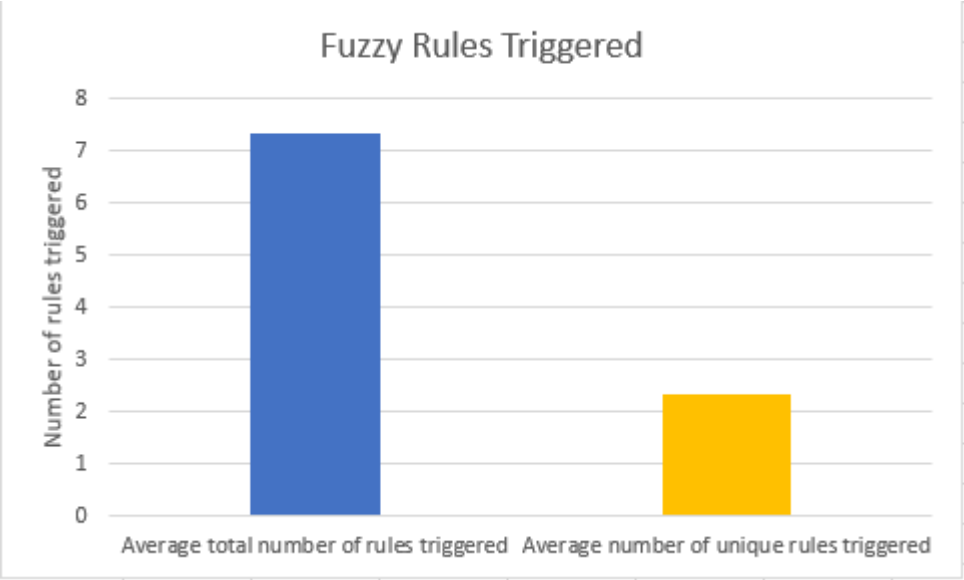

Figure 23 – (Fuzzy Logic) Number of total and unique rules triggered

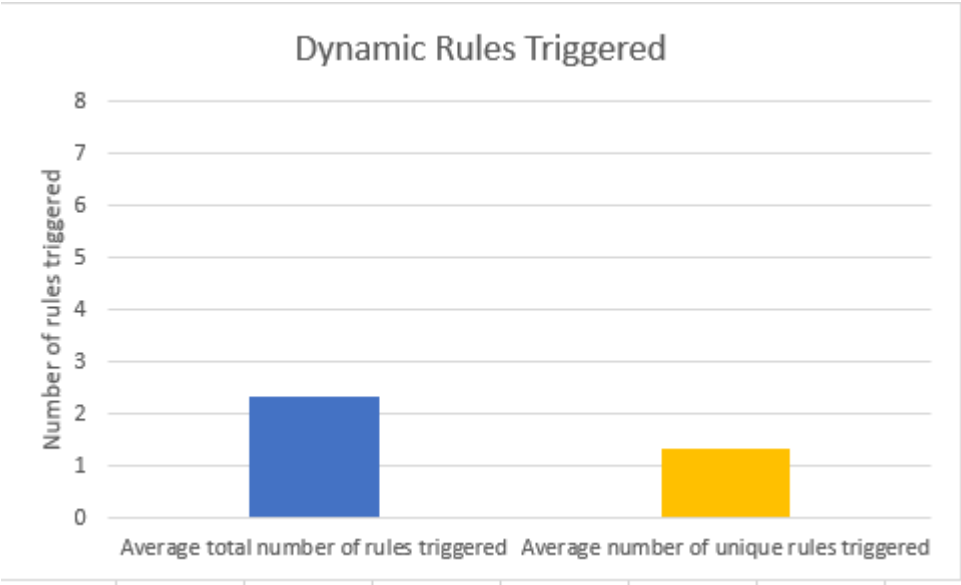

Figure 24 – (Dynamic Scripting) Number of total and unique rules triggered

The last 2 graphs from figures 23 and 24 present the average number of rules and unique rules triggered during the testing of both dynamic scripting and fuzzy logic. The fuzzy logic system, overall triggered a wider range of rules and more frequently.

5. Discussion

This section will analysis and evaluate the data that was presented in the results section. It will assess the differences between the fuzzy logic system and dynamic scripting system as a decision-making algorithm within the context of affective gaming. Also, to examine what the potential outcome could have been with a wider range of participants. Finally, this section will also discuss how the overall application compared to the initial research question and aims of the project.

As mentioned briefly in the previous section's introduction the total number of participants used for testing was 3. This was due to the current global circumstances where a national lockdown is in place. However, this will not stop the discussion of the potential outcome if there was a broader range of people to test the system with.

5.2 Fuzzy Logic Performance

The first area to examine is how effective the fuzzy logic system performed based on the physiological data recorded throughout the testing. Due to the nature of the game being a horror game, the general expectation is that there should be jumps in heart rate and decrease in valence, with the overall heart rate potentially higher towards the end of the game while valence potentially being lower at the end. Starting with the data seen in figure 16, the heart rate data remains relatively constant across all users with the only significant increases at around 8-9%, and the exception of user 2's considerable gain towards the end of the graph – it should be noted that user 2's data was kept to 120 samples as the data continue to record after the game was finished. An important note before evaluating the valence measurements recorded for the fuzzy logic is the potential issue of 'gamer face'. This refers to when people play or watch media on a screen where the user shows little to no emotions, this could affect the valence results as they are gathered via facial expression. From figure 17 the data shows that 2 of the 3 participants are hardly affected throughout the game with only small changes appearing, 'gamer face' maybe be one possible reason for this. The 3rd participant

being the exception with multiple spikes throughout the game – both peaks and troughs. From research done in section 2.2 negative valence suggests that the emotions associated with it are also negative such as sad, depressed, fearful, or annoyed. While observing user 2 it was noted that after certain in-game events they nervously laughed – which could potentially be read as positive valence by the software, explaining the peaks seen in figure 17. While overall the physiological data seen relatively constant with some exemptions, the pre and post SAM questionnaire (figure 12 & 13) suggests that there is some change in the user's mood. On average the user felt a notable increase in arousal (+2) and a slight decrease in valence (-0.66). This suggests that the overall impression of the fuzzy logic system seemed to have worked to some extent, by making the user on average fell less comfortable from decreased valence and arousal increase meaning the user is more on edge. Figures 20 to 22 display the 4 emotions recorded for each user during testing, the graphs are laid out as stacked area charts to show the emotional percentage trend over game time. From comparing the heart rate – which has a linear association with arousal – and the valence to the corresponding emotion on an affective model (figure 1), the outcome of each emotion within the fuzzy system seems to be calculated as excepted.

5.4 Dynamic Scripting Performance

With dynamic scripting, the same data was recorded apart from the individual emotions which are calculated by the FIS. When observing the data for heart rate for the dynamic scripting AI (figure 18), while the resting heart rate of each user is different, a similar trend can be seen with heart rate remaining somewhat constant across the participants. However, the spikes in heart rate are greater and more frequent, with increases between 16% and 22%. This primarily due to the data from user 3 which could potentially be seen as an outlier so, this data must be looked at sceptically. When examining the valence data recorded during the use of the dynamic scripting, there is a clear indication of negative

peaks of valence across the majority of users. Again, the 3rd user as the same in the heart rate data has extremely contrasting data when compared to the rest of the participants. A potential explanation for the high level of valence from user 3 seen towards the end of the graph, is that the 3rd user started to become frustrated by not being able to find the keys to complete the game, once found seem glad to finally find them. Finally, when viewing the pre and post SAM questionnaire for dynamic scripting, it shows a similar trend to that of the fuzzy logic, but not to the same extent. From figures 14 and 15 a minor increase in arousal and decrease in valence can be seen, with a +0.66 in arousal and -1.33 in valence. The greater decrease in valence compared to the fuzzy logic could be due to the irregular valence data from user 3. This leads to the same conclusion as the fuzzy logic system just but to a lesser degree, with physiological data remaining relatively constant throughout the playthrough, but user quantitative data from the SAM questionnaire suggests differently.

5.5 Questionnaire Results

When creating the questionnaire, the questions used were to gain qualitative feedback about the game, the AI systems, their opinions on affective gaming, and general information about the user. Of course, due to the extremely small sample size, this produces a limited range of opinions. Analysing the questions from figure 11 (full responses in appendix E), the first question asked to the user was a general question to gain if they are familiar or accustom to the horror genre, this would potential attain how this would then affect the user's later in terms of their physiological data. However, it could be argued that even the user's more familiar to the horror genre still have significant changes within their physiological data. To ensure that there was no bias when answer question 2, the systems were different for each participant. With 2 user's first testing the fuzzy logic, while the 3rd tested the dynamic scripting first. This can stop potential bias towards the 2nd AI system, if the user didn't find any of the systems impressionable overall and so choosing the latter

system as it is the most recent. While this switching of the systems technique was used during testing and fuzzy logic gaining 100% of the feedback, it is hard to comment if this trend would continue across a wider pool of participants. Question 3 was intended to assess if users were able to recognize that the AI systems were triggering rules based on their emotional status. With all users noticing in-game events in correlation to their emotions, however if the wording of the question was aimed at if the user thinks the AI notices their emotions and makes the appropriate decision, the outcome may be different. This simple change in wording in question 3 could have given greater feedback in relation to the research question for this project. The final 2 questions are more general questions about affective gaming overall. A potential issue associated with affective gaming is that the equipment required can be intrusive upon the user, creating an unpleasant experience. With the participants not finding the equipment intrusive, indicants that the choice of the webcam and ear-clip heart rate monitor was positive in not hindering the user's experience. The final question asked is to attain opinions from the participants on whether they believe affective gaming has a future within the games industry. This could indicate that further research should be conducted within the field of affective gaming if there is interest from users.

5.6 Comparison of Both AI Systems

This subsection will compare the results and data gathered from both systems. When looking at the limited data gain for heart rate and valence, both systems are relatively similar to the overtime trends mentioned before. However, the differences between the 2 systems appear when examining the SAM questionnaire data and the number of rules triggered data. When comparing the data sets from the SAM questionnaire, the fuzzy logic overall gives a greater increase in arousal than the dynamic scripting and a slightly less decrease in valence. While it seems the fuzzy logic system did a greater job of making the user feel more aware and on edge, with an increase in arousal. It can be seen that dynamic scripting

performed better at decreasing the user's valence, which is associated with more negative feelings. So, depending on how the game designer wants their AI to perform, it could be justified that both are suitable but for different outcomes, with fuzzy logic having a greater increase in data measurements overall. Figures 23 and 24 show the average total number of rules and unique rules triggered during testing. Initially, when viewing the graphs fuzzy logic has a clear distinction from dynamic scripting in terms of the average total number of rules triggered. Not only the average total number of rules but unique rules triggered too. This data can potentially explain the reason for all participants selecting the fuzzy logic system, as it seemed more engaging with more rules being activated during run-time. A potential reason for this disparity is the way the techniques are implemented, with the fuzzy logic rulebase having full access to all written rules during run-time. While dynamic scripting is constructed differently, where the algorithm selects a certain number of rules from the rulebase to run during gameplay. Only a new set of rules is selected once a rule is triggered, which means the dynamic scripting is limited in its choice in rules. Overall, the fuzzy logic system triggered on average 29.12% of the total rules designed, while dynamic scripting only triggered 16.62% of the total rules on average.

5.7 Project Analysis

While fuzzy logic seemed to perform better than dynamic scripting overall from testing with participants, this result must be taken with caution as mentioned due to the limited number of participants. Were testing done with a larger range of participants to test, the result could have been that fuzzy logic isn't the superior technique as first thought. While the results appear to display the project was successful to an extent, some areas could be improved.

Firstly, the game itself used to test the AI systems with participants is rather simplistic with basic game mechanics. While this could be argued that it makes the game straightforward and easy to test on a wide range of demographics, it makes the task of creating rules for the AI to analysis

and trigger more difficult. With more and greater game mechanics a larger rulebase could be constructed due to more possibilities.
Overall while the hardware used for this project seemed stable and adequate, further improvements in the choice of hardware could improve the experience as the AI systems would have access to more accurate and faster data. Questions could be asked about the accuracy of the chosen heart rate monitor for this project, while affordable the measurements taken during testing will not be to the same standard to that of medical-grade equipment. However, for the scale and scope of this project, the selected heart rate monitor seemed to be an appropriate choice. For the valence input device for the project – a webcam with Affectiva plugin - based on the data gathered during testing, potentially a different device should have been chosen such as electromyogram. This is due to the problem of ‘gamer face’ mentioned before where users express little to no emotions when playing or watching media.
The final area which could have been improved in relation to the research question and aim of the project is the questionnaire design. While the questions asked to the participants gave a basic understanding of their opinion towards the AI systems, most relative questions could have been asked to gain more feedback, also the potential of allowing open-ended questions rather than entirely closed-ended questions. Another option that could have been considered is the use of standardized questionnaires that have been used in the past by previous studies.

Overall the application that was created and tested provided elements of success towards the research question of the project from both AI systems, with fuzzy logic appearing to give better a result than dynamic scripting. However, as mentioned due to the current circumstances further testing with more participants is needed to gain a clearer result.

6. Conclusion and Future Work

6.1 Conclusion

The aim of this project was to answer the question “How effective are AI techniques with the use of affective sensors at improving player’s experience and immersion?”. While the results from the application don’t give a direct answer to this question, data and feedback gathered can give an indication that to some extent it did. Primarily from the fuzzy logic system.

From looking back at the initial research done before the development of the application, affective gaming can make use of multiple different types of decision-making algorithms, so that AI can make decisions based on the user’s physiological and psychological data. The literature review concluded that fuzzy logic and dynamic scripting both had the potential to work within affective gaming, from previous studies and research in similar fields.

The development of both the game and AI algorithms required significant time to implement with rearrangements of project planning needed to complete the project within scope and on time. Especially the fuzzy logic system which required a larger code structure than the dynamic scripting to function properly also needs a lot more research into the construction of membership functions and rules. Unfortunately, the rulebase designed and their respective actions ended up becoming limited with its diversity of potential actions.

While testing was limited to a small number of participants, data gathered seemed to indicate that out of the 2 systems implemented, fuzzy logic gave a better experience for the users playing. One of the potential issues recognized from testing is that the 5-minute break given to the participants between switch the AI system, this may have not been enough time for the human body to fully rest before starting the next playthrough of the game.

The conclusion taken from the project is that with the participants available, fuzzy logic provided the better experience and potential to

increase user immersion. Those that tested preferring the fuzzy logic system compared to that of the performance of the dynamic scripting, as well as all participants believing that affective gaming has a future within the games industry.

6.2 Future Work

Were the project to continue the following are just some suggestions and potential steps to take the project forward. Firstly, as mentioned before in the testing section and partly in the conclusion, is the fact that the number of participants for this project was extremely limited due to current circumstances. The first step would be to test the project again but with an increased pool of participants to gain a greater collection of data. Some potential alterations to be made before re-testing could include the restructure of the end of testing questionnaire. While the current questionnaire gains some basic feedback and data from participants, this is limited to a small number of questions and all being answered in closed-end responses. This could be expanded to include more questions about the user's experience of the game, potential issues with the gameplay, or rules that were triggered and allow for open-ended responses.

In terms of the actual game it's self this could be expanded with more game mechanics as mentioned earlier when analysing the project development. This could include anything from additional level design, more rules to be added to the rulebase with corresponding actions, greater use of the AI monster that was developed inside the game, or even potentially a small story/narrative to help immerse the player more in the environment of a haunted house.

Finally, the equipment itself could potentially be upgraded so that data is faster and more accurate. However, still ensuring that the equipment is within a reasonable price range for a consumer base interested in affective gaming. A potential switch of valence input device would likely give better valence readings. While the Affectiva plugin used is very effective, as cited before with 'gamer face', the valence data were few

and far between. Other potential devices to consider include electromyography, galvanic skin response, and electroencephalogram.

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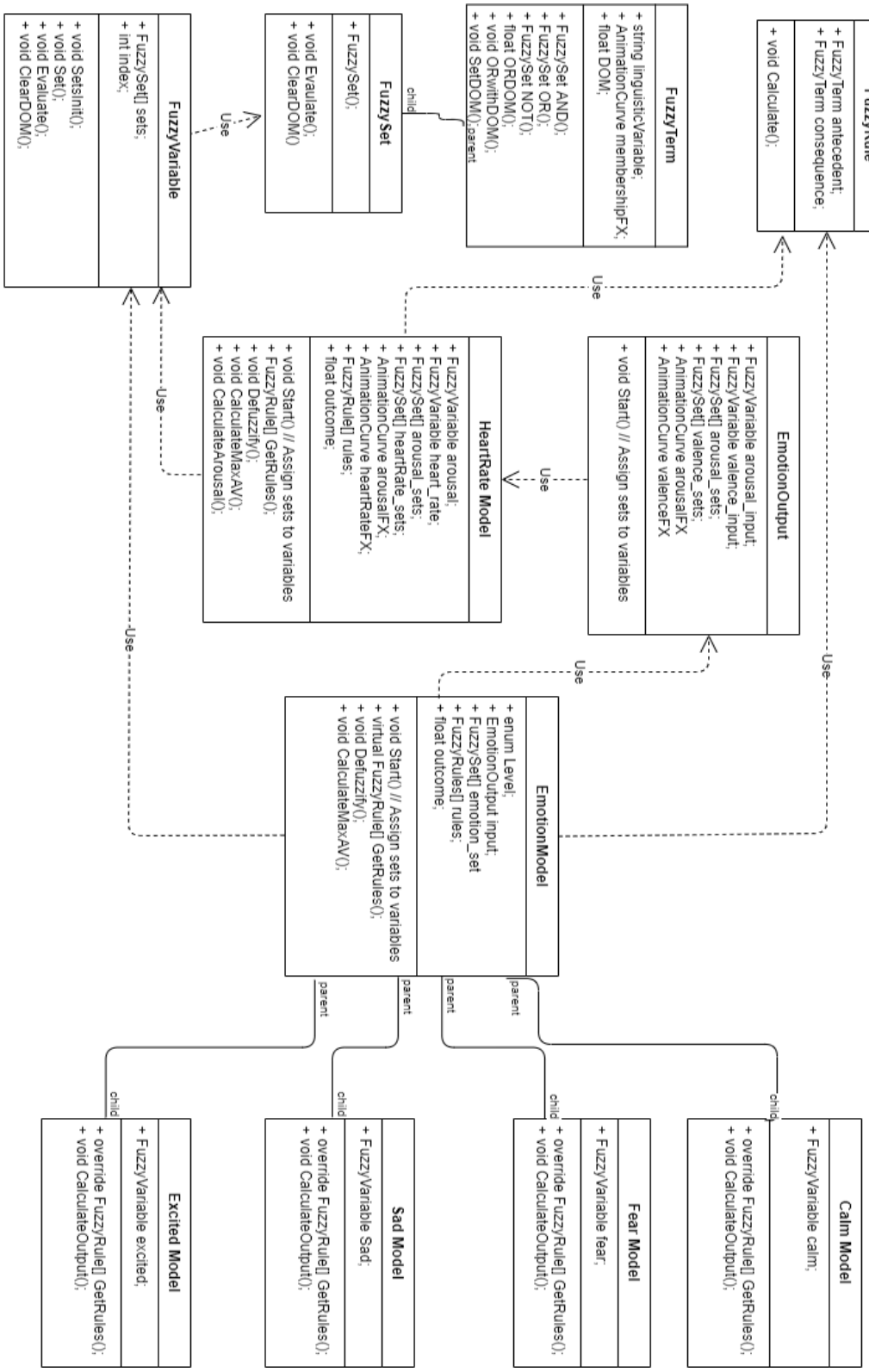
Appendices

Appendix A – Fuzzy logic class diagram



Appendix B.1 – Heart rate to arousal rules

Heart rate to arousal rules

IF (heart rate is low) THEN (arousal is low)

IF (heart rate is mid-low) THEN (arousal is mid-low)

IF (heart rate is mid-high) THEN (arousal is mid-high)

IF (heart rate is high) THEN (arousal is high)

Appendix B.2 – AV to emotion rules

Arousal-Valence to Calm rules

Arousal						
Valence		Low	Mid-low	Medium	Mid-High	High
	Low	Mid-Low	Low	Low	Low	Low
	Mid-Low	Medium	Mid-Low	Low	Low	Low
	Medium	Mid-High	Medium	Mid-Low	Low	Low
	Mid-High	High	Mid-High	Medium	Mid-Low	Low
	High	Mid-High	Medium	Mid-Low	Low	Low

Arousal-Valence to Excited rules

Arousal						
Valence		Low	Mid-low	Medium	Mid-High	High
	Low	Low	Low	Low	Low	Low
	Mid-Low	Low	Low	Low	Mid-Low	Mid-Low
	Medium	Low	Low	Mid-Low	Mid-Low	Medium
	Mid-High	Low	Mid-Low	Mid-Low	Medium	Mid-High
	High	Low	Mid-Low	Medium	Mid-High	High

Arousal-Valence to Sad rules

Arousal						
Valence		Low	Mid-low	Medium	Mid-High	High
	Low	Mid-High	High	Mid-High	Medium	Mid-Low
	Mid-Low	Medium	Mid-High	Medium	Mid-Low	Low
	Medium	Mid-Low	Medium	Mid-Low	Low	Low
	Mid-High	Low	Mid-Low	Low	Low	Low
	High	Low	Low	Low	Low	Low

Arousal-Valence to Fear rules

Arousal						
Valence		Low	Mid-low	Medium	Mid-High	High
	Low	Low	Mid-Low	Medium	Mid-High	High
	Mid-Low	Low	Mid-Low	Mid-Low	Medium	Mid-High
	Medium	Low	Low	Mid-Low	Mid-Low	Medium
	Mid-High	Low	Low	Low	Mid-Low	Mid-Low
	High	Low	Low	Low	Low	Low

Appendix C – Fuzzy Logic and Dynamic Scripting Rules

Fuzzy rule condition = Green highlight
Dynamic rule condition = Yellow highlight

// Jump Scare
IF Calm > 45 **THEN** Jump Scare with audio
IF HR < 45 **AND** Valence > 50 **THEN** Jump Scare with audio

// Spawn monster
IF Fear < 25 **AND** Excited < 25 **THEN** Spawn Monster
IF HR > 70 **AND** Valence > 50 **THEN** Spawn Monster

// Play eerie sound effect
IF Calm > 40 **AND** Sad > 40 **THEN** Play eerie sound effect
IF HR < 45 **AND** Valence >= 50 **THEN** Play eerie sound effect

// Play ambient sound effect
IF Calm < 50 && > 25 **AND** Sad > 25 && < 50 **THEN** Play ambient sound effect
IF HR < 50 **AND** Valence < 60 && > 30 **THEN** Play ambient sound effect

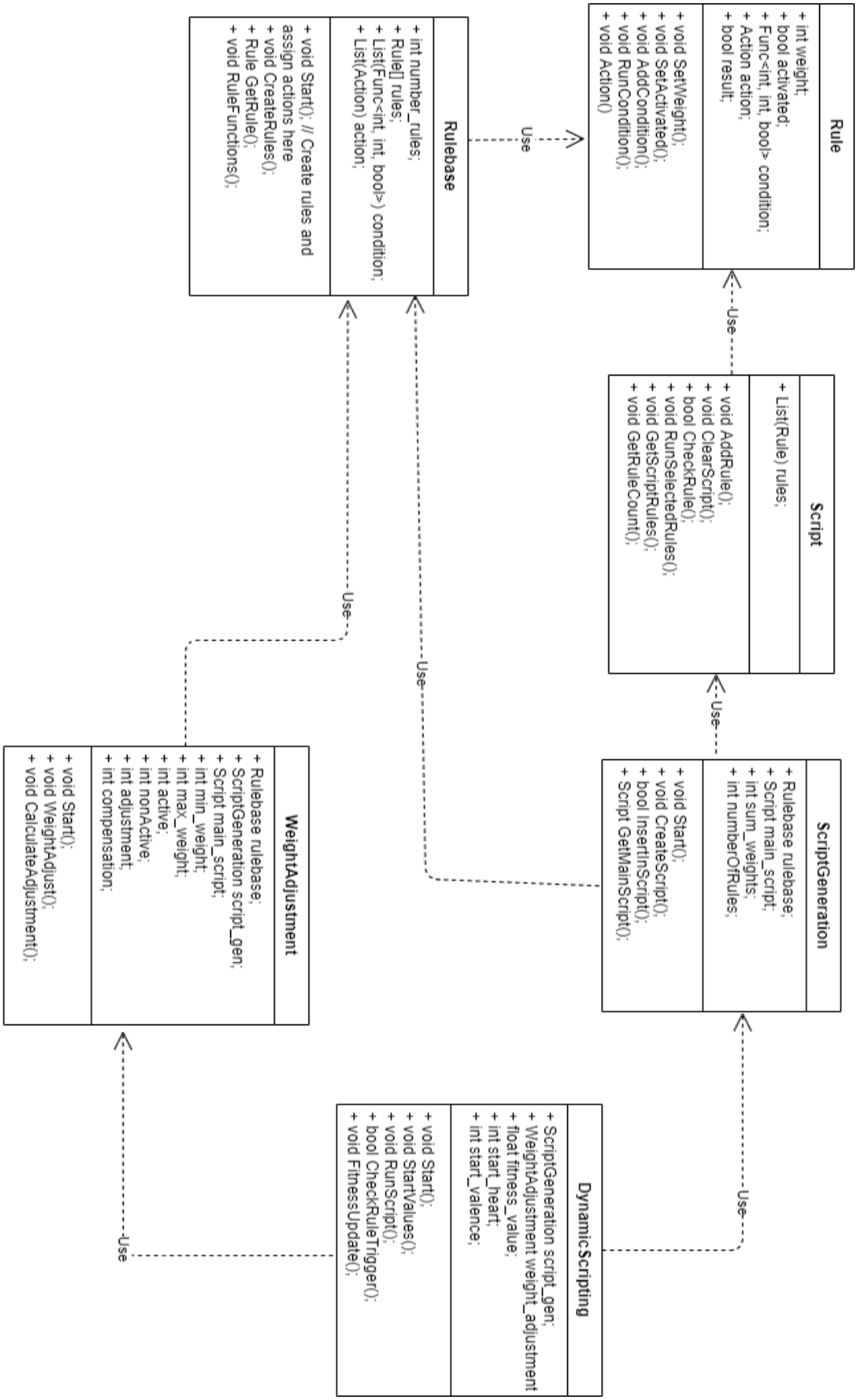
// Play scream sound effect
IF Excited > 45 **THEN** Play scream sound effect
IF HR > 65 **AND** Valence > 55 **THEN** Play scream sound effect

// Play evil laugh sound effect
IF Calm > 25 **AND** Sad > 25 **THEN** Play evil laugh sound effect
IF HR < 50 **AND** Valence < 50 **THEN** Player evil laugh sound effect

// Turn lights off
IF Fear < 20 **AND** Excited < 20 **THEN** Turn lights off
IF HR < 50 **AND** Valence <= 100 **THEN** Turn lights off

// Slow player movement
IF Excited > 50 **THEN** Slow player movement
IF HR > 70 **AND** Valence > 50 **THEN** Slow player movement

Appendix D – Dynamic scripting class diagram



Appendix E – End of testing questionnaire responses

		Yes	No
Q1)	Do you personally watch, read or play games within the horror genre?	1	2
		Fuzzy Logic	Dynamic Scripting
Q2)	Out of the 2 demos tested did you prefer the first or second?	3	0
		Yes	No
Q3)	Did you notice any in-game events (such as audio playing, lights turning off and on, or jump scares) in correlation to your emotions?	3	0
		Yes	No
Q4)	Did you find any of the equipment intrusive at all?	0	3
		Yes	No
Q5)	Do you believe affective gaming has a future within the video games industry?	3	0

Appendix F – GDPR Data Sign Off Form

Abertay University

GDPR Research Data Management
Data Sign Off Form

For undergraduate or postgraduate student projects supervised by an Abertay staff member.
This form MUST be included in the student's thesis/dissertation. Note that failure to do this will mean that the student's project cannot be assessed/examined.

Part 1: Supervisors to Complete

By signing this form, you are confirming that you have checked and verified your student's data according to the criteria stated below (e.g., raw data, completed questionnaires, superlab/Eprime output, transcriptions etc.)

Student Name:	Lewis Wilson		
Student Number:	1600546		
Lead Supervisor Name:	Jacqueline Archibald		
Lead Supervisor Signature	Jacqueline Archibald		
Project title:	Evaluation of AI Techniques using Affective Computing Solutions within Video Games		
Study route:	PhD ☐	MbR ☐	MPhil ☐
	Undergraduate ☑	PhD by Publication ☐	

Part 2: Student to Complete

	Initial here to confirm 'Yes'
I confirm that I have handed over all manual records from my research project (e.g., consent forms, transcripts) to my supervisor for archiving/storage	Yes
I confirm that I have handed over all digital records from my research project (e.g., recordings, data files) to my supervisor for archiving/storage	Yes
I confirm that I no longer hold any digital records from my research project on any device other than the university network and the only data that I may retain is a copy of an anonymised data file(s) from my research	Yes
I understand that, for undergraduate projects, my supervisor may delete manual/digital records of data if there is no foreseeable use for that data (with the exception of consent forms, which should be retained for 10 years)	Yes

Student signature : Lewis W

Date: 21/04/20